158.337 Database Development

Week 5 - Introduction to Structured Query Language (SQL)

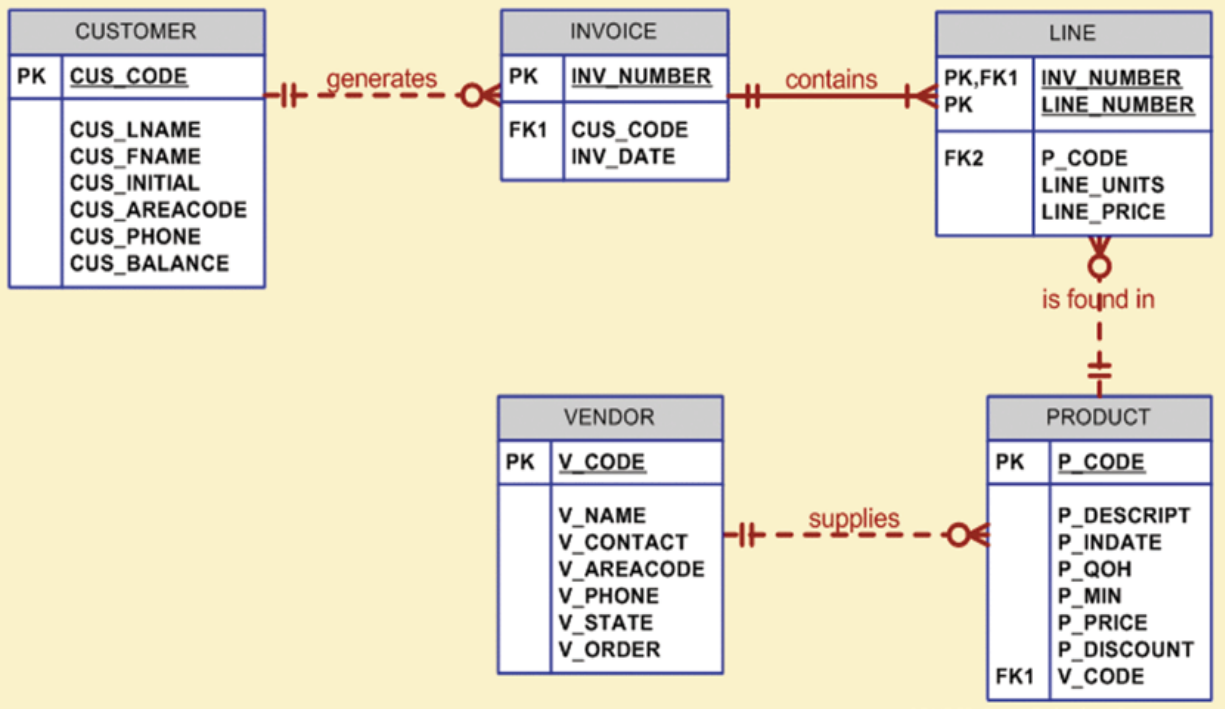
Read Chapter 7

SQL is a non-procedural language (i.e. specifies what is needed rather than how). SQL scripts are written for extracting/ manipulating/ re-structuring data. It has a vocabulary of about 100 commands – so not a lot of words – but LESS is MORE – and you will soon realize that SQL is a very powerful language for extracting information and managing databases.

There are several SQL dialects, but we will look at ORACLE only.

Business Rules for the case (Textbook)

1. A customer may generate many invoices. Each invoice is generated by one customer.
2. An invoice contains one or more invoice lines. Each invoice line is associated with one invoice.
3. Each invoice line references one product. A product may be found in many invoice lines
4. A vendor may supply many products. Some vendors do not supply products (yet to be approved)
5. If a product is vendor-supplied, that product is supplied by only a single vendor
6. Some products are produced in-house, i.e. they are not supplied by a vendor



Try out the scripts given in lecture slides and textbook.
These SQL queries mainly use the 2 tables – VENDOR and PRODUCT.

The scripts for creating these tables and inserting data are:

```
CREATE TABLE VENDOR (
 V_CODE     INTEGER     NOT NULL,
 V_NAME     VARCHAR(35) NOT NULL,
 V_CONTACT  VARCHAR(15) NOT NULL,
 V_AREACODE CHAR(3)     NOT NULL,
 V_PHONE    CHAR(8)     NOT NULL,
 V_STATE    CHAR(2)     NOT NULL,
 V_ORDER    CHAR(1)     NOT NULL,
 PRIMARY KEY (V_CODE));

INSERT INTO VENDOR
        VALUES(21225,'Bryson, Inc.'    ,'Smithson','615','223-3234','TN','Y');
INSERT INTO VENDOR
        VALUES(21226,'SuperLoo, Inc.'  ,'Flushing','904','215-8995','FL','N');
INSERT INTO VENDOR
        VALUES(11111,'ABCDEFGHIJK', 'LMNOPQRST','222','222-2222','AA','Y');

CREATE TABLE PRODUCT (
 P_CODE     VARCHAR2(10)
   CONSTRAINT PRODUCT_P_CODE_PK PRIMARY KEY,
 P_DESCRIPT VARCHAR2(35) NOT NULL,
 P_INDATE   DATE         NOT NULL,
 P_QOH      NUMBER       NOT NULL,
 P_MIN      NUMBER       NOT NULL,
 P_PRICE    NUMBER(8,2)  NOT NULL,
 P_DISCOUNT NUMBER(5,2)  NOT NULL,
 V_CODE     NUMBER,
 CONSTRAINT PRODUCT_V_CODE_FK FOREIGN KEY (V_CODE) REFERENCES VENDOR);

INSERT INTO PRODUCT
        VALUES('23109-HB','Claw hammer' ,'20-JAN-2008', 23, 10,  9.95,0.10,21225);
INSERT INTO PRODUCT
        VALUES('SM-18277','1.25-in. metal screw, 25'   ,'01-MAR-2008',172, 75,  6.99,0.00,21225);
INSERT INTO PRODUCT
        VALUES ('BRT-345','Titanium drill bit', '18-OCT-07', 75, 10, 4.50, 0.06, NULL);
```

The lecture slides have been supplemented with many SQL scripts, to explain the implementation aspect for each topic. SQL is quite like spoken English, and you will easily understand the purpose of each script as you read them. But reading and understanding SQL is different from writing your own scripts. So, I suggest that you also try writing each script and testing them in SQL Developer.
We will be writing more SQL statements in the coming weeks.